

AI-Based Price Prediction System

A Synopsis
For
Mini Project

**BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE
& ENGINEERING**

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1. Introduction

The rapid advancement of Artificial Intelligence (AI) and Machine Learning (ML) has opened new possibilities in the field of real estate and architectural planning. Traditionally, architectural floor plan generation required expert knowledge, manual drafting, and significant time investment. Similarly, real estate price estimation depended on subjective assessments and limited market data analysis.

The proposed project, AI-Based Layout Planning and Price Prediction System, aims to design and develop an intelligent desktop application that combines AI-driven layout generation with machine learning-based property price prediction. The system allows users to input plot dimensions, number of rooms, and facing direction to automatically generate a 2D architectural floor plan and receive a predicted market price for the property.

The system is built using a PySide6 desktop frontend, a FastAPI backend, and two independent ML components: a LayoutModel for optimal room placement and a PricePrediction Model for property valuation. The architecture follows a clean three-tier client-server design that ensures modularity and scalability [1].

The platform provides the following core capabilities:

- AI-generated 2D floor plan from natural language or structured input
- ML-based real estate price prediction
- Layout constraint validation with auto-correction
- Vastu-compliant room placement suggestions
- Interactive desktop application with PDF export

The platform will help property developers, homeowners, and architects reduce planning time, optimize space utilization, and make data-driven investment decisions.

2. Literature Review

The integration of machine learning in architectural and real estate domains has been studied extensively in recent years. Researchers have explored both automated layout generation and property price estimation as distinct yet complementary problems.

LeCun et al. [1] demonstrated that deep learning models can learn spatial patterns and structural relationships, forming the theoretical foundation for AI-driven layout generation. Their work on convolutional neural networks showed that spatial data could be processed and generated with high accuracy.

Géron [2] provided comprehensive coverage of supervised learning techniques applicable to regression problems, including Random Forest and Gradient Boosting regressors that form the basis of the price prediction module in this project. Ensemble methods were shown to outperform single estimators on real estate datasets.

FastAPI documentation [3] demonstrates the advantages of Python-based asynchronous REST APIs for ML serving, enabling low-latency communication between the desktop frontend and the ML backend. This pattern has been widely adopted in production ML systems.

The reviewed literature establishes that combining a rule-based layout engine with an ML price predictor, delivered through a desktop application, represents a practical and effective approach to the problems of architectural planning and property valuation. However, few integrated systems exist that combine both capabilities in a single offline-capable desktop tool. This project aims to bridge that gap [4].

3. Problem Definition

Despite advances in AI and ML, the following challenges persist in the real estate and architectural planning domain:

1. Manual Layout Design - Architectural floor plan creation is time-consuming and requires expert knowledge.
2. Inaccurate Price Estimation - Property valuations are often subjective and lack data-driven support.
3. Lack of Vastu Compliance Tools - Most digital tools do not incorporate Vastu Shastra directional guidelines.
4. No Constraint Validation - Generated layouts are rarely validated against structural and spatial constraints.
5. Fragmented Toolset - Layout generation and price prediction exist as separate, disconnected applications.
6. Offline Requirement - Many AI tools require internet connectivity, limiting use in field conditions.

Therefore, there is a need for an integrated, offline-capable AI system that combines automated layout generation with ML-based price prediction, constraint validation, and Vastu-compliant room placement in a single desktop application.

4. Objectives

The primary objectives of the project are:

1. To design an AI-based system for automated 2D architectural floor plan generation.
2. To develop a machine learning model for real estate price prediction.
3. To implement a layout constraint validation and auto-correction mechanism.
4. To incorporate Vastu Shastra rules for directional room placement.
5. To build a FastAPI backend for efficient ML model serving.
6. To develop a PySide6 desktop application as the user interface.
7. To enable furniture placement and parking layout in generated floor plans.
8. To provide PDF export functionality for floor plans and price reports.
9. To ensure the system works offline without internet dependency.
10. To ensure scalability and modularity of the platform architecture.

5. Methodology

The methodology consists of multiple integrated stages covering input collection, ML processing, layout rendering, and result display:

5.1 Input Collection

The system collects the following user inputs through the PySide6 desktop UI:

- Plot dimensions (width and length in feet)
- Number of rooms (BHK type: 1BHK to 5BHK)
- Plot facing direction (North / South / East / West)
- Material grade (Economy / Standard / Premium)
- Parking requirement (Yes / No)
- Number of floors (G, G+1, G+2)

5.2 Layout Generation (ML Model)

The LayoutModel processes plot data to generate optimal room placement using rule-based algorithms combined with ML zone prediction:

- Plot rectangle analysis
- Optimal room placement using spatial algorithms
- Vastu direction-based room correction
- Staircase and corridor placement for multi-floor plans
- Furniture icon placement (bed, sofa, kitchen platform, WC)
- Car parking placement with entry arrow

5.3 Price Prediction (ML Model)

The PricePrediction Model estimates property value using:

- Built-up area calculation from room dimensions
- Material grade cost multipliers
- Location zone-based pricing factors
- Regression-based ML model (Random Forest / Gradient Boosting)
- Cost breakdown by category: structure, finishing, plumbing, electrical

5.4 Constraint Validation

Generated layouts are validated against the following constraints:

- Minimum room size requirements
- Structural adjacency rules (e.g., kitchen not adjacent to bedroom)
- Vastu compliance score calculation
- Auto-adjustment if validation fails

5.5 Backend API (FastAPI)

The FastAPI backend exposes the following REST endpoints:

- POST /generate-layout - Accepts plot data, returns layout JSON
- POST /predict-price - Accepts features, returns predicted price
- POST /validate-layout - Validates layout against constraints
- GET /health - API status check

5.6 Drawing Engine (Matplotlib)

The drawing engine converts room coordinate data into 2D architectural drawings:

- Room rectangles with color-coded types
- Room labels with dimensions
- North arrow and scale indicator
- Furniture icons scaled to room size
- Dimension lines in architectural style

5.7 User Interface (PySide6)

The desktop application provides:

- Input panel with plot size, room count, and direction controls
- Tabbed view for Ground Floor, First Floor, and Cost Summary
- Embedded Matplotlib canvas for floor plan display
- Price breakdown table

- Save and Download PDF functionality

5.8 System Architecture

The architecture follows a three-tier design:

PySide6 Frontend UI (Qt Widgets + Matplotlib) → FastAPI Backend (ML Model Server) → ML Components (LayoutModel + PricePrediction Model) → JSON Results → Display on UI

6. Expected Outcomes

The platform is expected to:

1. Generate accurate 2D floor plans from user input within seconds.
2. Predict real estate prices with acceptable ML model accuracy ($R^2 > 0.85$).
3. Produce Vastu-compliant layout suggestions for all facing directions.
4. Validate and auto-correct layout constraint violations.
5. Provide downloadable PDF reports combining floor plan and cost estimate.
6. Operate fully offline without internet connectivity.
7. Reduce floor plan design time from hours to seconds.
8. Support multi-floor residential plan generation (G, G+1, G+2).

7. Tools and Technologies Used

- Python 3.10+ (Core development language)
- PySide6 / Qt6 (Desktop GUI application framework)
- FastAPI (Python REST API backend framework)
- Matplotlib (2D floor plan drawing engine)
- Scikit-learn (Machine learning models for price prediction)
- Shapely (Geometric operations for room placement)
- SQLite (Local database for storing layouts and predictions)
- ReportLab (PDF generation for floor plan export)
- spaCy / NLTK (Natural language processing for text input)
- Git / GitHub (Version control)

8. Future Scope

The system can be extended to:

- 3D visualization of generated floor plans using Blender or Three.js
- Cloud-based deployment for web browser access
- Mobile application version for Android and iOS
- Integration with CAD export formats (DXF / DWG) for professional use
- Multi-city real estate price prediction with location zone datasets
- AI-driven interior design suggestions
- Real-time collaborative floor plan editing
- Integration with government land record APIs for plot verification
- Multi-language NLP support (Hindi, Gujarati input prompts)

9. Conclusion

The AI-Based Layout Planning and Price Prediction System represents a significant step towards intelligent, data-driven architectural planning and real estate decision-making. By integrating a PySide6 desktop interface, a FastAPI backend, and two independent ML components —

LayoutModel and PricePrediction Model — the system delivers a unified, offline-capable platform for generating floor plans and estimating property prices.

The project demonstrates how artificial intelligence and machine learning can automate traditionally manual, expert-dependent processes. The Vastu-compliant layout generation, constraint validation, and cost estimation features make the platform practically useful for homeowners, developers, and architects in the Indian real estate context.

The modular three-tier architecture ensures that each component — UI, API, and ML — can be independently updated and extended, providing a strong foundation for future enhancements including 3D visualization, cloud deployment, and mobile applications.

10. References

Journal References

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